

General Characteristics

61. A transition element X has a configuration $[Ar]3d^4$ in its + 3 oxidation state. Its atomic number is
 (a) 25 (b) 26
 (c) 22 (d) 19
62. The transition element have a characteristic electronic configuration which can be represented as
 (a) $(n-2)s^2p^6d^{1-10}(n-1)s^2p^6ns^2$
 (b) $(n-2)s^2p^6d^{1-10}(n-1)s^1p^6d^1$ or d^1ns^1
 (c) $(n-1)s^1p^6d^{10}ns^2p^6nd^{1-10}$
 (d) $(n-1)s^2p^6d^{1-10}ns^1$ or ns^2
63. Number of unpaired electrons in Fe^{+++} ($Z = 26$) is
 (a) 4 (b) 5
 (c) 6 (d) 3
64. Of the ions Zn^{2+} , Ni^{2+} and Cr^{3+} [atomic number of $Zn = 30$, $Ni = 28$, $Cr = 24$]
 (a) Only Zn^{2+} is colourless and Ni^{2+} and Cr^{3+} are coloured
 (b) All three are colourless
 (c) All three are coloured
 (d) Only Ni^{2+} is coloured and Zn^{2+} and Cr^{3+} are colourless
65. Common oxidation state of scandium, a transition element is/are [atomic number of $Sc = 21$]
 (a) + 4 (b) + 1
 (c) + 2 and + 3 (d) + 4 and + 1
66. Which of the following is not correct about transition metals
 (a) Their melting and boiling points are high
 (b) Their compounds are generally coloured
 (c) They can form ionic or covalent compounds
 (d) They do not exhibit variable valency
67. From +6 to +1 oxidation state is shown by the element of..... group
 (a) V-B (b) VI-B
 (c) VII-B (d) VIII
68. The electronic configuration of cobalt is
 (a) $1s^2, 2s^2 2p^6, 3s^2 3p^6 3d^1, 4s^2$
 (b) $1s^2, 2s^2 2p^6, 3s^2 3p^6 3d^7, 4s^2$
 (c) $1s^2, 2s^2 2p^6, 3s^2 3p^6 3d^3, 4s^2$
 (d) $1s^2, 2s^2 2p^6, 3s^2 3p^6 3d^5, 4s^2$
69. Out of all the known elements the number of transitional elements is
 (a) 80 (b) 61
 (c) 43 (d) 38
70. **Cigarette** or **gas lighter** is made up of
 (a) Misch metal
 (b) Alkali metal
 (c) Noble metal
 (d) None
71. **Bullet-proof steel** alloy is prepared by using
 (a) Sc (b) Ni



- (c) *Zr* (d) *Zn*
72. In making **gun-berrel**, the steel used is
(a) *Fe – Mn* (b) *Fe – Cr*
(c) *Fe – W* (d) *Ni – Mo*
73. Neobium and tantalum metals are used in making surgical instruments because they are
(a) Non-corrosive (b) Hard
(c) Soft (d) All
74. To support tungsten filament in electric bulb, the steel used is
(a) *Cr* (b) *Ni*
(c) *Mn* (d) *Mo*
75. The elements belonging to the first transition series have atomic number from
(a) 19 to 37 (b) 22 to 32
(c) 24 to 30 (d) 21 to 30
76. Which of the following elements does not belong to the first transition series
(a) *Fe* (b) *V*
(c) *Ag* (d) *Cu*
77. Fe^{2+} shows
(a) Ferromagnetism
(b) Paramagnetism
(c) Diamagnetism
(d) None of these
78. Zinc and mercury do not show variable valency like *d*-block elements because
(a) They are soft
(b) Their *d*- shells are complete
(c) They have only two electrons in the outermost subshell
(d) Their *d*-shells are incomplete
79. Cuprous ion is colourless while cupric ion is coloured because
(a) Both have half filled *p* and *d*-orbitals
(b) Cuprous ion has incomplete *d*-orbital and cupric ion has a complete *d*-orbital
(c) Both have unpaired electrons in the *d*-orbitals
(d) Cuprous ion has a complete *d*-orbital and cupric ion has an incomplete *d*-orbital
80. Transition metals are related to which block
(a) *s*-block
(b) *p*-block
(c) *d*-block
(d) None of these

